RED SEAL EXAM PREP

309A Construction Electrician

Free 50-Question Mock Exam

Practice for the Red Seal / Certificate of Qualification exam

Pass mark: 70% · Aligned to the published occupational standard topic weightings
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Mock Exam — answer all questions, then check the key

Q1. The stated object of the Canadian Electrical Code, Part I (Section 0) is to establish safety standards for the installation and maintenance of electrical equipment, with consideration given to the prevention of:

A) Excessive voltage drop on branch circuits and feeders

B) Inefficient use of energy in building electrical systems

C) Interference with communication and signalling systems

D) Fire and shock hazards to persons and to property

Q2. When pulling conductors through conduit, what is the maximum allowable conductor fill percentage for a conduit containing three or more conductors?

A) 60% fill — commercial conduit systems allow higher fill than residential

B) 50% fill — half the conduit interior area may be used by conductors

C) 31% fill — applies to two conductors in the same conduit

D) 40% fill — the limit for three or more conductors

Q3. In a reversing motor starter, why must the forward and reverse contactors be interlocked?

A) To transfer overload protection between the two contactors

B) To stop both contactors from closing at the same time

C) To let the motor coast to a stop before it reverses

D) To limit inrush current when the motor changes direction

Q4. A ground fault occurs on a 120 V branch circuit protected by a 15 A breaker. The fault current is only 3 A — far too low to trip the breaker. What device would protect a person against this fault?

A) A larger 20 A breaker, since a higher rating responds to smaller faults

B) A fuse of the same rating, because a fuse is faster on a small fault

C) A Class A ground fault device, which trips on 4 to 6 mA in milliseconds

D) An arc fault device, which detects any fault below the breaker rating

Q5. In a three-phase system, the relationship between line voltage (V_L) and phase voltage (V_Ph) in a WYE (star) connection is:

A) $V_L = V_{Ph} \div \sqrt{3}$

B) $V_L = V_{Ph}$ (equal)

C) $V_L = V_{Ph} \times 3$

D) $V_L = V_{Ph} \times \sqrt{3}$

Q6. **A single-phase load draws 10 kVA at a power factor of 0.8 lagging. How much reactive power does it draw, and what would a correction capacitor have to supply to bring the power factor to unity?**

A) 8 kVAR, and a capacitor supplying 8 kVAR would correct it

B) 6 kVAR, and a capacitor supplying 6 kVAR would correct it

C) 2 kVAR, being the shortfall between the 10 kVA and 8 kW

D) 6 kW, since reactive power is measured in kilowatts too

Q7. **What is the minimum approach distance for an unqualified worker near an energized overhead power line at 25kV in Canada?**

A) 10 metres

B) 5 metres

C) 1 metre

D) 3 metres

Q8. **A VFD (Variable Frequency Drive) controls motor speed by:**

A) Varying the motor's pole count electronically

B) Switching between star and delta windings

C) Varying both frequency and voltage proportionally

D) Varying the supply voltage only, keeping frequency constant

Q9. **A buck-boost transformer is connected as an autotransformer to lift a 208 V supply to about 230 V for a motor. What does that connection give up compared with a two-winding transformer of the same rating?**

A) The ability to change voltage, which needs two separate windings

B) Overcurrent protection, which a two-winding transformer provides

C) Isolation — the two circuits share a winding and a conductor

D) Efficiency, because an autotransformer loses more in its core

Q10. **In an Ontario house the metal gas piping serves a water heater and a range, neither of which has any electrical connection. The piping has been bonded with a No. 6 AWG copper conductor to a ground rod driven outside beside the gas meter, and that rod connects to nothing else. What does the Ontario Electrical Safety Code require here?**

A) Nothing further, since gas piping falls to the fuels authority

B) The bonding conductor must be No. 4 AWG copper rather than No. 6

C) The piping must be made equipotential to the electrical system

D) The rod is acceptable if its resistance to earth is low enough

Q11. **An apprentice trained in the United States quotes a National Electrical Code article to justify an installation on a Canadian job. What actually governs the work?**

A) The manufacturer's instructions, which override any code wording

B) The National Electrical Code, which is recognized across Canada

C) Whichever of the two codes is the more demanding on that point

D) The provincial electrical code, being the CE Code as adopted there

Q12. Under the CEC, what is the maximum voltage drop allowed in a branch circuit for power and heating loads?

A) 5% maximum for all circuits — no distinction between feeder and branch circuit drop

B) 3% for the branch circuit and 5% total from service to point of utilization

C) Voltage drop is not regulated by the CEC — it is a design guideline only

D) 1% maximum — power loads are more sensitive to voltage drop than lighting

Q13. What is the purpose of a bonding conductor in an electrical installation?

A) Bonding provides the return path for current during normal operation — it carries load current

B) It ties all metal enclosures into a low-impedance fault path so devices trip quickly

C) Bonding prevents static electricity buildup — it is only required in flammable/explosive locations

D) Bonding is another term for the neutral conductor — they serve the same function

Q14. A 6 m straight run of 1-inch EMT has been fastened only at the outlet boxes at each end, with no intermediate supports. Why must an inspector reject it?

A) Each strap doubles as the bonding connection the tubing needs to stay grounded

B) The unsupported span sags and puts its load on the couplings and connectors

C) EMT may not be run in a straight length greater than 3 m without a pull box

D) Straps are what let the tubing expand and contract, so the run will buckle

Q15. A series AC circuit has a resistance of 12 ohms, an inductive reactance of 16 ohms and a capacitive reactance of 7 ohms. What is the impedance?

A) 15 ohms, using Pythagoras on 12 ohms and 9 ohms of net reactance

B) 21 ohms, from adding 12 ohms to the 9 ohms of net reactance

C) 35 ohms, from adding the resistance and both reactances together

D) 20 ohms, using Pythagoras but ignoring the capacitive reactance

Q16. Under the CEC, what clearance must be maintained between an overhead service entrance conductor and the finished grade of a residential property?

A) 6 metres above all outdoor areas regardless of vehicle access

B) 3 metres above grade at all points

C) 3.5 m over pedestrian areas, 4 m over residential driveways, 5.5 m over roads

D) Clearance is only specified for power lines over 600V — residential service has no CEC height requirement

Q17. A motor winding is tested with a megohmmeter for ten minutes at one steady test voltage. The reading is 8 megohms at one minute and 8.5 megohms at ten minutes. What does that flat result indicate?

A) A winding too warm for a valid test

B) A fault forming between winding and frame

C) Moisture or dirt in the winding

D) Insulation that is dry and sound

Q18. **In the Canadian Electrical Code, what is a "wet location", and what does that classification require?**

A) A location subject to saturation with water - a drip shield fitted above each enclosure satisfies the Code

B) A location that is underground or buried only - direct-burial cable is the one wiring method the Code permits

C) A location where liquids may drip, splash or flow on equipment - wet-rated wiring and weatherproof enclosures

D) A location where moisture may condense on or near equipment - ordinary dry-location wiring methods are accepted

Q19. **A single dwelling unit of 140 square metres is being serviced. Among its loads are an electric range rated 12 kW and a 5 kW clothes dryer. Under the Canadian Electrical Code, what does the range contribute to the calculated load used to size that service?**

A) 15 000 W, since the Code adds 25% to the nameplate rating

B) 3000 W, since the Code takes a range at 25% of nameplate

C) 6000 W, the demand the Code fixes for a range that size

D) 12 000 W, since the Code carries a range at full nameplate

Q20. **Transformer nameplates give a rating in kVA rather than kW. Why is the rating expressed that way?**

A) Because its limits are heat and flux, which ignore the load's power factor

B) Because the rating already includes the transformer's own internal losses

C) Because kVA is simply the older unit, and kW would state the same limit

D) Because transformers deliver only reactive power, which is measured in kVA

Q21. **A technician is installing Type AC (armoured cable) and needs to connect it to a metal box. What is required at the connection point?**

A) The armour must be soldered to the box for a continuous ground path

B) An AC-approved connector plus an anti-short (red) bushing over the cut armour

C) A grounding lug must be attached externally to the armour before entering the box

D) The armoured cable can be connected directly to the box knockout — no fittings required

Q22. **A three-phase induction motor is running but makes a loud hum and vibrates excessively. One phase has been lost (single-phasing). What happens to the motor?**

A) It keeps running but overheats from high current and reduced torque

B) The motor reverses its direction of rotation when single-phasing

C) The motor runs normally at slightly reduced speed and efficiency

D) The motor stops at once and will not restart when one phase is lost

Q23. **A three-phase motor draws high current on all three phases but runs at reduced speed under load. The MOST likely cause is:**

A) Single-phasing — one supply line open

B) Low supply voltage on all three phases

C) Overload relay set too high for the motor

D) Motor oversized for the application it drives

Q24. **CSA Z462 requires a distribution panel to be placed in an electrically safe work condition before maintenance begins. What has to be true of the panel for that condition to exist?**

A) Wearing arc flash PPE rated for the incident energy available at the equipment

B) De-energizing, locking out, releasing stored energy, proving absence of voltage

C) Posting a worker at the disconnect so that no one can close it during the work

D) Switching the main breaker off and confirming that the panel loads have stopped

Q25. **A feeder conductor has an ampacity of 75 A from the applicable table. After the correction for the number of current-carrying conductors sharing its raceway, its allowable ampacity is 60 A. Under the CEC, what is the largest overcurrent device permitted to protect that conductor?**

A) 60 A, the allowable ampacity of the conductor as installed

B) 30 A, half the conductor allowable ampacity as a margin

C) 48 A, eighty percent of the conductor allowable ampacity

D) 75 A, the ampacity of the conductor as listed in the table

Q26. **A 600 V three-phase motor is running hot. The three line-to-line voltages at its terminals read 600 V, 588 V and 576 V, and the three line currents read 22 A, 18 A and 18 A. What do these readings point to?**

A) Shorted turns in the winding on the 22 A line

B) Normal operation for a loaded three-phase motor

C) A failing breaker on the line carrying 22 A

D) A supply voltage unbalance of about 2%

Q27. **The reactance of a capacitor (X_C) in an AC circuit:**

A) Increases as frequency increases

B) Decreases as frequency increases

C) Is not affected by frequency

D) Is equal to the capacitance value in farads

Q28. **A wound rotor induction motor drives a rock crusher. Its rotor windings are brought out through slip rings to a bank of external resistors. How is that resistance normally used?**

A) The slip rings work as a commutator, so the resistors set field current as they would on a DC motor

B) DC fed through the slip rings pulls the rotor into step with the stator field near full speed

C) The resistors remain permanently in circuit to hold running speed below synchronous under load

D) Full resistance is in circuit at start, then cut out in steps as the motor comes up to speed

Q29. **Under the Canadian Electrical Code, which of the following is evidence that a piece of electrical equipment is approved for installation in Canada?**

A) A mark from a certification body accredited by the Standards Council of Canada

B) A CE mark, which shows the product meets the European safety standards in force

C) A CSA mark specifically, since marks from other agencies are not accepted here

D) A test report from the manufacturer, kept on file for the inspector to review

Q30. **What does EMT stand for in electrical conduit systems?**

A) Electrical Metallic Tubing

B) Electrical Mechanical Threading

C) Enclosed Metal Trough

D) Exterior Metal Tube

Q31. **A 120/208 V three-phase, four-wire feeder is run in No. 2 AWG copper and is colour-coded. Which colours does the CE Code call for on phases A, B and C?**

A) Phase A red, phase B white, phase C blue

B) Phase A black, phase B red, phase C blue

C) Phase A red, phase B black, phase C blue

D) Phase A orange, phase B brown, phase C yellow

Q32. **What is the purpose of the overload relay in a motor starter?**

A) To protect the circuit conductor from overcurrent conditions

B) To prevent the motor from starting in reverse direction

C) To protect the motor against short-circuit fault current

D) To protect the motor windings from sustained overload

Q33. **What does LOTO stand for and when must it be applied?**

A) Lock Out, Tag Out — only for high voltage above 600V

B) Line Output Test Operation — during electrical testing

C) Load Only, Turn Off — when changing light bulbs

D) Lockout/Tagout — before maintenance or service on equipment

Q34. **A VFD (Variable Frequency Drive) is installed on a pump motor and the operator reports the motor makes a high-pitched whine but operates correctly otherwise. What is the most likely explanation?**

A) The VFD's PWM carrier frequency is in the audible range — a normal characteristic

B) The VFD is malfunctioning — high-pitched noise indicates an output fault

C) The motor bearings are failing — a high-pitched whine is early bearing failure

D) The pump impeller is cavitating due to low suction pressure

Q35. **What distinguishes a three-phase synchronous motor from a squirrel cage induction motor in normal running?**

A) It holds synchronous speed with no slip, because its rotor carries DC field excitation

B) It runs slightly faster than synchronous speed, by an amount that depends on the load applied

C) It develops full starting torque from standstill without any auxiliary starting arrangement

D) It slips further below synchronous speed than an induction motor because its rotor has no cage

Q36. **An AC circuit has R = 8 Ω and XL = 6 Ω in series. What is the total impedance?**

A) 4.8 Ω (8 × 6 ÷ 10)

B) 14 Ω (8 + 6)

C) 2 Ω (8 − 6)

D) 10 Ω ($\sqrt{8^2 + 6^2}$)

Q37. **A Type NMD90 cable is being installed in a residential building. According to the CE Code, in which locations is NMD90 cable installation NOT permitted?**

A) NMD90 may not be used in finished basements — only conduit is allowed there

B) NMD90 may be installed anywhere in a residential building without restriction

C) In wet locations, in concrete or masonry, or exposed to mechanical damage

D) NMD90 is not permitted in exterior walls only — all interior walls are fine

Q38. **A capacitor bank is switched in across a motor load and the measured line current falls noticeably, while the machinery keeps doing exactly the same work. What has happened?**

A) The capacitors now supply the reactive current the motor needs

B) The capacitors reduce the real power the motor draws from the line

C) The capacitors lower the supply voltage, so less current is drawn

D) The capacitors store energy and release it back into the machinery

Q39. **Before drilling through a wall to run conduit in an existing building, a technician MUST:**

A) Assume the wall is clear if no outlets are visible on the surface

B) Drill a test hole first to check for studs

C) Notify the building manager and wait 24 hours

D) Verify the location of hidden wiring, plumbing, and gas lines

Q40. **A three-phase motor runs in the wrong direction. What is the SIMPLEST correction?**

A) Swap any two of the supply phase conductors

B) Rewind the motor stator windings

C) Replace the motor starter contactor

D) Add a starting capacitor to the circuit

Q41. **A technician is working alone in an electrical room. Under Canadian occupational health and safety regulation, lone-worker requirements typically include:**

A) No special requirements - electrical work can always be done alone

B) Working alone requires only notifying a supervisor before starting

C) Lone work is prohibited outright for any electrical task

D) A check-in system, buddy system, or remote monitoring

Q42. **A 20-amp branch circuit must use conductors rated for at least:**

A) 30 amps — one size up for safety

B) 15 amps — one size down is permitted

C) 25 amps — to allow for load growth

D) 20 amps — equal to the device rating

Q43. **According to CEC Table 53, what is the minimum cover for a direct-buried armoured cable such as TECK90, rated 750 V or less, in an area not subject to vehicular traffic?**

A) 150 mm, because the armour is the protection that matters

B) 450 mm, reducible by 150 mm where protection is added

C) No minimum — armour lets the cable be laid at any depth

D) 900 mm, the same for every direct-buried cable and depth

Q44. **A circuit has a power factor of 0.75 and draws 20A at 240V (single phase). What is the real power (watts) consumed?**

A) 2,400W

B) 3,600W

C) 6,400W

D) 4,800W

Q45. **When running EMT conduit through a fire-rated wall assembly, what is required at the penetration?**

A) Nothing — EMT is metal and inherently fire resistant

B) A junction box on each side

C) An approved firestop system at the penetration

D) A locknut on each side of the wall

Q46. **Why does a metal raceway carrying a consumer's service conductors need bonding connections that a feeder raceway does not?**

A) A fault there is ahead of the service overcurrent device

B) The raceway serves as the grounding electrode for the service

C) Service conductors sit at a higher voltage than feeders

D) The main breaker is slow to clear a fault of that size

Q47. **On an Ontario construction project, under what circumstances does the construction regulation permit work on exposed energized parts instead of disconnecting and locking out first?**

A) The panel carries an arc flash label and the worker's arc-rated clothing is rated above the incident energy printed on it

B) Disconnecting is not reasonably possible, would create a greater hazard at 600 V or less, or the work is diagnostic testing

C) The worker holds a certificate of qualification as an electrician and wears rubber gloves with leather protectors over them

D) The supervisor in charge of the project authorizes the entry in writing and a competent worker stands by to observe the job

Q48. Under the CEC, what is the purpose of a ground fault protection device fitted to a large solidly grounded service?

A) To protect against lightning strikes on the service entrance

B) To detect ground faults before arcing destroys the switchgear

C) To provide GFCI (shock) protection for personnel

D) To balance the load between phases

Q49. An existing bathroom receptacle beside the wash basin in an older Ontario house has failed and is being replaced like for like, with no other change to the circuit. What does the Ontario Electrical Safety Code require of the replacement?

A) A breaker at the panel rather than a device at the outlet

B) Class A ground fault protection on the replacement device

C) A plain replacement, since the rule is not retroactive here

D) Arc fault protection, which is what a wet area calls for

Q50. A contactor coil is rated 120VAC. When measured with a multimeter, the coil reads 0Ω (zero resistance). What does this indicate?

A) Normal — coil resistance is always near zero

B) The coil is shorted — it will draw excess current

C) The coil is open — it will not energize

D) The coil is functioning correctly

Answer Key & Explanations

Q1. Answer: D

The Code's stated object is safety: the prevention of fire and shock hazards. Section 0, Object, states that the Code establishes safety standards for the installation and maintenance of electrical equipment, where consideration has been given to the prevention of fire and shock hazards, and that its requirements address the fundamental principles of protection for safety of IEC 60364-1. Section 2, General Rules, does not carry that statement: its technical rules cover marking of equipment, guarding for the protection of persons and property, maintenance and working space, and enclosure selection for the environment. Voltage drop, energy use and signal interference are dealt with by particular rules or by other documents; none of them is the Code's stated object.

Q2. Answer: D

Three or more conductors in one conduit: 40% maximum fill. The fill limits exist so conductors can be pulled without stripping insulation and so the assembly can still shed heat. One conductor on its own is allowed 53%, because a single conductor is pulled straight and jams nothing; two conductors drop to 31%, the tightest of the three, because two round conductors can wedge side by side against the conduit wall; three or more return to 40%, since a bundle of three or more tends to twist together rather than jam. Fill is worked out by area, not by counting: total cross-sectional area of the conductors including their insulation, divided by the internal cross-sectional area of the conduit. The Code carries the fill percentages in its conduit and tubing fill table, with separate tables of conductor cross-sectional areas and of conduit internal areas to do the arithmetic with.

Q3. Answer: B

Interlocking prevents a line-to-line short circuit. A reversing starter uses two contactors feeding the same motor, with two of the three line conductors swapped between them. If both sets of main contacts closed together, those two line conductors would be tied directly to each other - a phase-to-phase fault across the supply, cleared only by the branch overcurrent device. Reversing starters therefore carry a mechanical interlock (a physical lever or bar that blocks the second armature) plus an electrical interlock (a normally closed auxiliary contact on each contactor wired in series with the opposite coil), and often a pushbutton interlock as well. The relay does not manage inrush, coasting, or overload protection: a single overload relay sits downstream of both contactors and serves either direction.

Q4. Answer: C

A Class A ground fault device compares the current leaving on the ungrounded conductor with the current returning on the identified conductor and trips on a difference of 4 to 6 mA, which is roughly a thousandth of what this breaker needs. Three amperes through a person is not a survivable current — around 100 mA through the chest can stop a heart — yet a 15 A breaker will sit there indefinitely, because from its point of view a three-ampere load is simply a small load. That gap between what kills a person and what a breaker notices is the entire reason ground fault protection exists. Reading the wrong answers: an arc fault device is watching the waveform for the signature of arcing and is fire protection, so a clean ground fault through a person passes it unremarked; fitting a larger breaker moves the trip point further away, not closer; and a fuse of the same rating has the same problem the breaker does, since it is also sized to the conductor rather than to a person. Test the device with its own test button on a schedule — the electronics can fail with the receptacle still delivering power, so an untested device is an assumption rather than a protection.

Q5. Answer: D

Wye: V_Line = V_Phase × √3. In a wye connection, each phase connects between one line and neutral. Line voltage (between two lines) = phase voltage × 1.732. Example: 208V wye system has phase voltage = 208/1.732 = 120V. Delta connections: V_Line = V_Phase (equal).

Q6. Answer: B

Real power is 10 kVA times 0.8, or 8 kW. The three quantities form a right triangle, so the reactive power is the square root of 10 squared minus 8 squared, which is the square root of 36, or 6 kVAR — and a capacitor supplying 6 kVAR cancels it. The triangle is the point. Apparent power is the hypotenuse, real power the horizontal side and reactive power the vertical side, and they combine at right angles because the reactive current is 90 degrees out of phase with the working current. Reading the wrong answers: taking 8 as the reactive figure confuses the two sides of the triangle and gives the real power a second job; subtracting 8 from 10 treats the sides as though they added arithmetically, which they never do; and quoting a reactive quantity in kilowatts loses the distinction the units exist to preserve, since watts measure power that does work and volt-amperes reactive measure power that does not. Correcting to unity in one step is the textbook answer rather than the field answer: correction is normally sized to a target such as 0.95, because a bank that overcorrects drives the power factor leading, can raise the voltage at light load, and may resonate with harmonics already on the system.

Q7. Answer: D

25kV = 3 metre minimum approach distance. Provincial occupational health and safety regulations set minimum approach distances by voltage, and the first high-voltage band is 3 metres across the country: Ontario 3m from 750 to 150,000 volts, Alberta 3.0m from 750V to 40kV, British Columbia 3m from over 750V to 75kV. A 25kV distribution line sits inside that band in every province. Below 750V the recommended distance is 1 metre. Qualified line workers work to different limits set by their training, procedures and PPE.

Q8. Answer: C

VFD: varies frequency and voltage proportionally (V/Hz ratio). Motor speed depends on supply frequency. A VFD converts AC to DC then back to variable-frequency AC. Voltage is reduced proportionally with frequency to maintain the same magnetic flux (torque) at all speeds. This allows smooth, efficient speed control from 0–60Hz+.

Q9. Answer: C

An autotransformer has one winding with a tap rather than two windings, so the input and the output share turns and share a physical conductor: there is no isolation between the supply and the load. A two-winding transformer keeps its primary and secondary electrically separate and couples them only magnetically, which is what lets the secondary be grounded independently and what stops a fault or a disturbance on one side from appearing directly on the other. Take that away and the two circuits are one circuit with a tap on it. Reading the wrong answers: efficiency is a strength of the autotransformer rather than a weakness, since only the difference between input and output is transformed and the rest is passed through the shared winding, which is exactly why it is smaller, cheaper and more efficient for a modest voltage change; changing voltage is precisely what an autotransformer does, and it does it with one winding; and no transformer of either kind provides overcurrent protection, which comes from a separate device sized to the conductors. The consequence to carry into the field is the one that matters for grounding and bonding: because there is no isolation, an autotransformer does not create a separately derived system, so the neutral is not re-bonded at it and the grounding arrangement of the supply carries straight through. A two-winding transformer does create one, and its derived neutral is bonded to a grounding electrode at the source, at a single point.

Q10. Answer: C

The piping has to be made equipotential to the building's own electrical system, and a rod that connects to nothing else does not do that. Rule 10-700 requires the metal gas piping of a building supplied with electric power to be made equipotential, meaning at a substantially equal electric potential, to the non-current-carrying conductive parts of electrical equipment. The Electrical Safety Authority puts this exact arrangement to itself in its bulletin on equipotential bonding of non-electrical equipment and answers it plainly: installing a new ground electrode connected to the gas system does not meet the rule, because what the piping must be made equipotential to is the system grounding conductor, not a separate isolated ground electrode. The reason is the word equipotential. A stand-alone electrode gives the piping its own reference to earth, so it can sit at a different voltage from the panel, the raceways and the appliance cases around it, and that difference is what a person bridges. Land the conductor on the bonding bus in the panel instead and pipe and enclosures rise and fall together; where a building ends up with more than one grounding electrode, Rule 10-702 requires the electrodes to be interconnected for the same reason. Reading the wrong answers: a low resistance reading at the rod rescues nothing, because the rule asks what the piping is at the same potential as rather than how well it reaches earth, and the Authority's answer carries no resistance qualifier; the fuels authority does not relieve anyone of this, since Technical Safety BC states that equipotential bonding of non-electrical systems including gas piping is regulated electrical work, enforced under the electrical permit, although the caution runs the other way as well and satisfying the electrical code does not by itself satisfy everything the gas regulator asks; and the conductor already run is the right size, because the minimum for exposed wiring not subject to mechanical damage is No. 6 AWG copper or No. 4 AWG aluminum, No. 4 being the aluminum figure and not the copper one. Two limits travel with this rule. Metal gas piping threaded into a gas-fired appliance whose electrical supply contains a bonding conductor is already equipotential through that supply, which is why the appliances in this house are specified as having no electrical connection. And piping sections interconnected by corrugated stainless steel tubing are not required to be jumpered together, that tubing being bonded for lightning protection under the fuels regulator and the manufacturer's instructions.

Q11. Answer: D

What governs is the electrical code the province has adopted into law, which is the Canadian Electrical Code, Part I, together with that province's own amendments. The CE Code is a standard published by CSA Group; it acquires legal force only where a province or territory adopts it by regulation, and provinces adopt different editions on different dates and add amendments of their own. Ontario's Electrical Safety Code, for example, is the CE Code with Ontario amendments, administered by the Electrical Safety Authority; British Columbia's is administered by Technical Safety BC. Reading the wrong answers: the National Electrical Code has no legal standing anywhere in Canada, and citing it to an inspector will not save an installation; picking whichever code is stricter is not how adoption works, and in practice the two differ in kind rather than in strictness, since Canadian ground fault protection is triggered by distance from water where the American rule lists rooms; and manufacturer's instructions matter for the equipment but do not displace the code. The practical consequences are constant: UF cable, THHN insulation and the American ampacity tables have no Canadian equivalent to cite, and equipment used in Canada must bear the certification mark of a body accredited for Canada.

Q12. Answer: B

Rule 8-102: 3% maximum in a feeder or a branch circuit, and 5% maximum overall from the supply side of the consumer's service to the point of utilization. The two limits stack — a feeder may use up to 3%, the branch circuit it supplies may use up to 3% more, but the two together may not exceed 5% at the furthest outlet. In Canada these figures are mandatory: the rule says "shall". The US NEC recommends the same 3% and 5% figures, but only in informational notes, so there they are advisory rather than enforceable — that is the real difference between the two codes, not the numbers. Excessive voltage drop makes a motor draw more current to hold its torque, which overheats it, and it dims lighting and shortens equipment life.

Q13. Answer: B

Bonding creates a low-impedance fault current path to ensure rapid OCPD operation during a ground fault. Without bonding, a fault from a live conductor to a metal enclosure would result in the enclosure becoming energized at line voltage — dangerous touch voltage. With proper bonding, fault current flows freely back to the source through the bonding/grounding system, causing the fuse or breaker to operate within milliseconds and clearing the fault.

Q14. Answer: B

Support is a mechanical requirement. Tubing that hangs between two boxes carries its own weight, the weight of the conductors in it, and any vibration in the structure, and all of that load ends up in the couplings and the box connectors at each end. Those joints work loose, the tubing eventually pulls out of a connector, and the raceway stops being continuous and mechanically protected. Section 12 of the code therefore requires EMT to be fastened at intervals along the run and again close to each box, fitting or termination; read the actual interval from the edition in force in your jurisdiction, since it is not a single number for every trade size. Straps are not bonding devices - bonding continuity comes from the tubing and its fittings - and they are not expansion fittings, which are a separate fitting used where the building itself moves.

Q15. Answer: A

Inductive and capacitive reactance oppose each other, so they subtract first: 16 minus 7 leaves 9 ohms of net inductive reactance, and the impedance is the square root of 12 squared plus 9 squared, which is the square root of 225, or 15 ohms. The order matters. Current lags the voltage across an inductor and leads it across a capacitor, so on the impedance diagram the two reactances point in opposite directions and cancel down to whatever is left over. Only the surviving reactance is then combined with the resistance, and that combination is at right angles, which is why it takes Pythagoras rather than addition. Reading the wrong answers: adding all three values gives 12 plus 16 plus 7, or 35 ohms, and treats reactance as though it were resistance; keeping Pythagoras but dropping the capacitor gives the square root of 144 plus 256, or 20 ohms, which is what happens when a capacitor is read as just another element in the string rather than an opposing one; and adding 12 to the correct 9 ohms of net reactance gives 21 ohms, the right first step spoiled by the wrong second one. Because the inductive reactance is the larger of the two, this circuit is still net inductive and the current lags. Had the two been equal the net reactance would be zero, the impedance would fall to the resistance alone and the current would be at its maximum, which is series resonance and the reason a power factor correction capacitor is sized rather than simply added.

Q16. Answer: C

CEC Rule 6-112(3) overhead conductor clearances: 3.5 m over pedestrian-only areas, 4 m over residential driveways, 5 m over commercial driveways, 5.5 m over roads and lanes. Overhead service entrance conductors must maintain minimum vertical clearances above finished grade to prevent accidental contact. A residential driveway requires 4 m; areas where trucks and commercial vehicles pass (commercial driveways, roads, lanes) require 5 m and 5.5 m respectively. Check CEC Rule 6-112 for complete clearance requirements by installation type.

Q17. Answer: C

The reading barely moved across the ten minutes, and that flat curve is the signature of moisture or conductive dirt: the ten-minute reading divided by the one-minute reading - the polarization index - is 8.5 over 8, about 1.06. Clean, dry insulation behaves in a particular way when a steady DC test voltage is held on it. The absorption and polarization currents decay as the minutes pass, the total leakage falls, and the resistance the instrument reports climbs, often several times over. Insulation carrying moisture or conductive contamination supports a steady leakage current that does not decay, so the reported resistance sits about where it started. The reason the test is held for ten minutes is that it asks about the shape of the curve rather than the height of it, so a winding whose single reading looks acceptable can still be caught here; the response is to dry and clean the winding and test it again before the machine is trusted. Reading the wrong answers: dry, sound insulation is what produces a rising reading, so a flat one is the warning rather than the reassurance; a fault forming between the winding and the frame collapses the reading toward zero rather than holding it steady at megohms; and temperature moves the size of the readings, which is why a single insulation resistance value is corrected for temperature before it is compared with anything, but a warm winding is no reason for the reading to stop climbing. Two cautions go with the test. Where the one-minute reading is already very high there is almost no absorption current left to decay, so the ratio becomes unreliable and is read alongside the single resistance value rather than instead of it. And the motor is disconnected from any variable frequency drive before a megohmmeter is applied to it.

Q18. Answer: C

Wet location: liquids may drip, splash or flow on or against the electrical equipment. The Canadian definition turns on liquid reaching the equipment. "Subject to saturation" is the wording of the US National Electrical Code and is not the Canadian test. A damp location is one normally or periodically subject to condensation of moisture in, on or next to the equipment, including partly protected places under canopies, marquees and roofed open porches. A wet location calls for wet-rated wiring methods - TECK90 or ACWU90, NMWU where the run is direct-buried, or raceway with the entries sealed - and, as Technical Safety BC's bulletin on equipment exposed to the weather puts it, equipment exposed to splashing water must be of a weatherproof or watertight type, while equipment exposed only to falling or condensing moisture may be drip-proof, weatherproof or watertight. Car washes, spray areas and outdoor locations are the usual examples.

Q19. Answer: C

The Code assigns a single electric range rated up to 12 kW a demand of 6000 W, whatever the nameplate says inside that band. Section 8 does not add up nameplates. The calculated load for a single dwelling is built from a basic load for floor area, a fixed demand for the range, the greater of the space-heating and air-conditioning loads where an interlock prevents both from running at once, and the remaining loads brought in at a reduced factor - because a house never runs everything at full output at the same moment, and a service sized as though it did is a service nobody can afford. Reading the wrong answers: carrying the full 12 kW forward is exactly the error the demand factor exists to prevent, and on a 240 V supply that one appliance would inflate the calculated load by 25 A; adding a margin on top of the nameplate borrows a continuous-load style multiplier that the demand rules do not apply here; and 25% is the factor for the other loads, the dryer in this example and a storage water heater alongside it, not for the range. The discipline is to classify each load before doing any arithmetic - a range demand, a floor-area basic load and an other load are three lines with three different treatments, and a load written on the wrong line changes the service size. A range rated above 12 kW does not stay at 6000 W: the demand rises by 40% of the amount by which the rating exceeds 12 kW, so a 14 kW range is 6800 W.

Q20. Answer: A

The two things that limit a transformer are the heat its winding current produces and the flux its applied voltage produces, and neither one knows or cares what the power factor of the load is. Winding loss is I squared R, set by the current alone; core loss is set by the voltage and frequency, which fix the flux. Multiply the rated voltage by the rated current and you have volt-amperes, which is the honest statement of what the machine can carry. Rate it in kilowatts instead and the figure would be wrong for every load except a purely resistive one: a 100 kVA transformer feeding a load at 0.7 power factor delivers 70 kW while its windings carry exactly the same current, and exactly the same heat, as they would at 100 kW into a resistive load. Reading the wrong answers: a transformer delivers whatever mixture of real and reactive power the load demands and is not restricted to either; the choice of unit is not historical convention but a statement about what the limit is; and losses are accounted separately, since the rating describes output capability rather than a net figure with losses subtracted. The same reasoning explains why generators and uninterruptible supplies are rated in kVA and why a load schedule adds volt-amperes rather than watts — and it is the reason a badly corrected plant can run its transformers to their limit while the meter shows real power well below the nameplate.

Q21. **Answer: B**

AC cable connection: approved connector + anti-short bushing required. When cutting armoured cable, the spiral steel armour leaves sharp edges that can cut conductor insulation. The anti-short bushing (red plastic bushing) is inserted between the cut armour edge and the conductors. An approved AC connector clamps the cable to the box. The armour and internal ground wire (if present) provide the grounding path.

Q22. **Answer: A**

Single-phasing: the motor keeps running but draws dangerously high current and overheats. With one line open the rotating field collapses into a pulsating field, which decomposes into a forward and a backward rotating component. The motor keeps turning on its momentum and on the forward component, but its torque is reduced and now pulsates, which is what produces the hum and the vibration. The two remaining lines carry the whole load current, typically 1.5 to 2 times normal full-load current, and the windings overheat quickly unless the overload relay trips.

Q23. **Answer: B**

Low voltage + slow speed + high current = under-voltage condition. When supply voltage drops, motors slip more to develop the same torque, drawing higher current. Check supply voltage at the motor terminals under load. Also check for high resistance connections causing voltage drop.

Q24. **Answer: B**

An electrically safe work condition is something a worker creates and then proves; it is not a precaution taken around equipment that is still live. Ontario's Construction Projects regulation sets out, for work on electrical equipment, installations and conductors, that the power supply be disconnected, locked out of service and tagged, that hazardous stored electrical energy be adequately discharged or contained before the work begins, and that the worker verify before starting that both have been done. That is the shape of the condition: the supply gone and held gone, the energy already inside the equipment dealt with, and the result proved by measurement rather than assumed. Verification is the step no lock can stand in for. A lock proves a handle cannot be moved; it says nothing about a back-fed circuit, a second supply, a control transformer fed from another panel, or a switch blade that failed to open. So absence of voltage is measured at the conductors with an adequately rated instrument, and the instrument is proved on a known live source before the test and again after it, because an instrument that failed between the two would read zero on a live panel. Where the installation calls for it, temporary protective grounds go on once absence of voltage has been proved. Reading the wrong answers: arc flash PPE is worn precisely because the panel is still being treated as live, so it protects the worker from the hazard instead of removing it; watching the panel loads stop tells you that current stopped flowing through them and nothing about what is still live behind the cover; and a worker posted at the disconnect can be called away or overruled, which is why each person exposed to the hazard applies a lock of their own. The Canadian Electrical Code has a say here as well, since Rule 2-304(1) permits no repairs or alterations on live equipment except where complete disconnection is not feasible, while the detailed work procedure comes from CSA Z462 and from the occupational health and safety regulation covering the job.

Q25. Answer: A

An overcurrent device may not be rated above the allowable ampacity of the conductor it protects, and allowable ampacity means the corrected figure for the conductor as it is actually installed. The table value assumes conditions this raceway does not meet, so the conductor cannot carry 75 A here. What it can carry is 60 A, and that is where the ceiling on the device sits. Reading the wrong answers: taking the table figure leaves the conductor guarded by a device that will not open until well past the current the installation actually permits, and that is the failure this rule exists to prevent, because the wire and its terminations overheat while the breaker sees nothing worth tripping on. The eighty percent figure is real, but it belongs to a different question: it limits how much continuous load an ordinary circuit may carry, so 48 A is the continuous load this conductor may serve, not the largest device that may protect it, and confusing a load limit with a device rating leaves a circuit that cannot carry what it was installed for. Halving the ampacity is invented restraint, since nothing in the Code cuts a conductor to 30 A to choose its protection, and it strands capacity the conductor genuinely has. Watch the trap in the other direction as well: the allowance to move up to the next larger standard rating exists only where the allowable ampacity falls between two standard device ratings, and it is not a general licence to round a corrected ampacity upward. Correct the conductor first, then size the device to what is left.

Q26. Answer: D

The supply itself is measurably unbalanced, so the ammeter is reporting a consequence and not a cause. Work the voltages first. They average 588 V, the greatest departure from that average is the 12 V at either end, and 12 divided by 588 is 2.0%. Now the currents: they average 19.3 A, the greatest departure is the 2.7 A on the high line, and that is about 14%. A three-phase induction motor turns a voltage unbalance into a current unbalance several times its size, so a supply out by 2% readily produces a current spread of this order - the two readings tell one consistent story, and the heat comes with it, because the negative-sequence component an unbalanced supply introduces meets a very low impedance in the rotor and generates heat out of all proportion to the size of the unbalance. Reading the wrong answers: a spread of this size is not what a healthy installation looks like, and treating it as normal is how motors are cooked; the winding is not implicated by these readings, because everything on the ammeter is already accounted for by the supply, and a winding is condemned only after the supply has been corrected and the currents measured again; and a breaker carries whatever current the motor draws rather than setting it, so changing it would alter nothing. Chase the unbalance back through the feeder - single-phase loads spread unevenly across the three phases, a loose or corroded termination in one line, or unbalance arriving from the utility - and derate the motor until it is fixed.

Q27. Answer: B

Capacitive reactance decreases with increasing frequency. $X_C = 1/(2\pi fC)$. As frequency rises, the capacitor charges and discharges more rapidly, offering less opposition to current flow. At DC (0Hz), X_C approaches infinity (blocks DC). At very high frequencies, X_C approaches zero (passes easily).

Q28. Answer: D

External rotor resistance is a starting aid: all of it in circuit at standstill, shorted out in steps as the motor accelerates. Resistance in the rotor circuit raises rotor power factor and moves the point of maximum torque toward standstill, so the machine develops high breakaway torque while drawing far less line current than a squirrel cage motor started across the line. That is why wound rotor motors suit crushers, hoists and other high-inertia, high-breakaway loads. A secondary controller short-circuits the resistance in steps as speed builds, and at full speed the rings are shorted and the machine runs with the characteristics of a squirrel cage motor. Leaving resistance in does give a lower, load-sensitive speed, but with poor speed regulation and all the lost power dissipated as heat in the resistor bank, so it is not the normal running condition. The slip rings carry AC induced in the rotor windings; they are not a commutator, and no DC excitation is applied, which is what a synchronous machine does.

Q29. Answer: A

Approval is a certification body's finding, and the body's mark on the equipment is the evidence of it. The bodies that count are those accredited by the Standards Council of Canada, and the equipment must have been certified against a recognized Canadian standard. Where a piece of equipment carries no such mark - a one-off machine, a rebuilt assembly, an import - it is not approved, and it is not made approved by an inspector looking at it on site; the route open to it is a field approval by the provincial safety authority or another accredited organization, which ends in an approval label of its own. Reading the wrong answers: the CE mark is the manufacturer's own declaration against European requirements and carries no recognition in Canada, which is exactly why imported equipment so often has to be field approved before it can be energized; CSA is one accredited certification body among more than twenty, and the marks of the others are equally good, so treating "CSA approved" as the definition of approved is trade shorthand rather than the rule; and a manufacturer's test report is not a certification at all, because nothing independent stands behind it.

Q30. Answer: A

EMT = Electrical Metallic Tubing (thin-wall conduit). Metallic, not "Metal" — that is the exact wording of the Canadian product standard, CSA C22.2 No. 83 "Electrical metallic tubing" (steel version: CSA C22.2 No. 83.1), and of the Canadian Electrical Code subdivision that covers it, Section 12, Rules 12-1400 to 12-1414. EMT is lightweight and non-threaded, and is joined with set-screw or compression fittings instead of threads. Rule 12-1402 sets the permitted uses: exposed and concealed work, and since the 2015 CE Code wet and outdoor locations as well, provided wet-location connectors and couplings are used and a bonding conductor is run in wet-location tubing. Before any underground run, check Rule 12-012, Underground installations, for the permitted raceway types and cover depths. The tubing gives the conductors mechanical protection and, when properly bonded, forms part of the equipment bonding path.

Q31. Answer: C

Where circuits are colour-coded, the CE Code marks three-phase ac insulated conductors phase A red, phase B black and phase C blue, with a white neutral. The order matters more than it looks, because the one competing scheme in Canadian practice differs precisely at phase B: licensed distributors, licensed transmitters and licensed generators may mark their high voltage installations phase A red, phase B white or yellow, phase C blue, with a bare or concentric neutral. Wherever consumer conductors meet supply authority conductors there are then two conventions in one enclosure, and a phase conductor marked white that lands on a neutral lug in a service box is a shock and equipment failure waiting to happen. Reading the wrong answers: swapping the first two phases is the habit carried over from single-phase work, where black is the first line conductor; red, white and blue is the supply authority scheme quoted outside the high voltage context it belongs to; and orange, brown and yellow is permitted by Section 24 for isolated systems in patient care areas, from where some shops have extended it to 347/600 V circuits to tell them apart from 120/208 V, but the provincial guidance is that new installations use red, black and blue exclusively and that raceways, junction boxes and panelboards be identified instead with the voltages they contain. In the size given, the identified conductor is a continuous white covering or three continuous white stripes along the entire length of the conductor, and an insulated bonding conductor has a continuous green finish or green with one or more yellow stripes.

Q32. Answer: D

The overload relay protects the motor itself from current above full load but far below a fault level - the current a branch-circuit breaker or fuse will carry all day without noticing. Short-circuit and ground-fault protection is the branch-circuit overcurrent device's job. The overload relay's job is the slow thermal damage caused by a jammed load, a lost phase, a failed bearing or a long run at excess current. Its time delay lets the motor draw its normal high starting current without tripping, but it trips on sustained excess current before the winding insulation cooks. Reading the wrong answers: the relay is set from the motor nameplate full-load current, so the motor is what it is chosen to protect, and it cannot serve as the conductor's overcurrent device in any case, because a thermal element working through a contactor cannot interrupt fault current - which is why a fuse or breaker is still required ahead of every starter; a short circuit is cleared in a fraction of a cycle by that fuse or breaker, orders of magnitude faster than any thermal element can move; and direction of rotation is set by the phase sequence at the motor terminals, which is a wiring question rather than a protection one. Thermal relays, bimetallic or eutectic-alloy, must cool before they can be reset, while an electronic relay senses the current directly and carries adjustable trip points; either type may be arranged for manual or for automatic reset. The setting itself comes off the motor nameplate: read the full-load amperes, then take the maximum percentage of that current from the edition of the Code in force where you work.

Q33. Answer: D

LOTO = Lockout/Tagout. Required any time a worker could be injured by unexpected startup or release of stored energy (electrical, hydraulic, pneumatic, gravity). LOTO applies to ALL voltage levels — even 120V can kill.

Q34. Answer: A

VFD motor whine: the PWM carrier frequency sits in the audible range. A VFD rectifies the incoming AC to DC and then switches it back to AC by pulse-width modulation. The rate at which the output transistors switch — the carrier frequency — sets the frequency of the magnetic noise the motor radiates, and on most drives the factory setting falls well inside the band a person can hear, which reaches to roughly 20 kHz. Raising the carrier shifts the tone toward the top of that band, where it is far less obtrusive and inaudible to many adults; it does not move the noise above human hearing. Raising it also costs more switching loss and heat in the drive's output stage, and puts greater voltage stress on the winding insulation of a motor that is not inverter-duty rated, so the setting is always a compromise. Reading the wrong answers: cavitation is a low gravelly rattle heard at the pump itself, not a steady electrical tone; a drive with a genuine output fault trips rather than running the pump correctly; and a failing bearing grows louder and hotter with running time and shows up on a vibration or temperature check, while carrier whine is constant and present from the first start.

Q35. Answer: A

A synchronous motor locks to the rotating field: shaft speed is 120 x f divided by poles at every load within its rating. Its rotor is not a shorted cage relying on induced current. It carries a DC-excited field, fed through slip rings, that is pulled into step with the stator field. A load increase changes the torque angle between rotor and stator field, not the speed, so the shaft holds exactly synchronous speed until the load exceeds pull-out torque, at which point the motor falls out of step and stalls rather than slowing gradually. Because the rotor field is stationary with respect to the rotor, a synchronous motor produces no net starting torque on its own: it is run up as an induction motor on damper (amortisseur) windings or by a drive, and the DC field is applied once it is near speed. An induction motor is the opposite case, and it can never reach synchronous speed, because without slip there is no induced rotor current and therefore no torque.

Q36. Answer: D

Impedance in series AC circuit: $Z = \sqrt{(R^2 + X^2)}$. Resistance and reactance are 90° out of phase, so they add as vectors, not arithmetic. $Z = \sqrt{(8^2 + 6^2)} = \sqrt{(64 + 36)} = \sqrt{100} = 10\ \Omega$. This is the Pythagorean theorem applied to phasors. Power factor = R/Z = 8/10 = 0.8 (lagging for inductive circuit).

Q37. Answer: C

NMD90 is a dry-and-damp-location cable: keep it out of wet locations, out of concrete and masonry, and protected wherever it can be damaged. The Electrical Safety Authority puts it plainly in Bulletin 12-19-17: "NMD90 is suitable for use in dry or damp locations only." Note that damp is not wet — NMD90 is lawfully run in bathroom walls and in basements. Wet means underground, in a raceway or slab in contact with earth, or otherwise exposed to weather; there the cable to use is NMWU, not NMD90. The same bulletin sets the protection rules a candidate is most likely to be marked on: where the cable passes through or runs along studs, joists or plates within 32 mm of the edge, an approved protection plate or bushing is required, and along a face it must be covered by corrosion-resistant ferrous metal not less than 1.3 mm thick. NMD90 may pass through a boxed-in return-air plenum where an approved insert protects it at the crossing, but it may not be fished lengthwise through one. The jacket is normally rated FT1, which is what keeps the cable out of buildings required to be of non-combustible construction.

Q38. Answer: A

The motor still needs the same magnetizing current; the capacitors are now supplying it locally instead of the utility supplying it down the whole length of the feeder. A motor draws two components of current: one in phase with the voltage that does the work, and one 90 degrees out of phase that builds and collapses the magnetic field without doing any work at all. A capacitor draws its current 90 degrees the other way, so placed alongside the motor it exchanges reactive energy with the motor cycle by cycle. The reactive current then circulates in the short loop between the two rather than travelling back to the transformer, and the line current falls to something much closer to the working component alone. Reading the wrong answers: the real power is unchanged, and it must be, since the machine is doing the same work and energy cannot be conjured by a capacitor; the supply voltage does not drop, and in fact correction usually raises the voltage slightly at the load by removing reactive current from the feeder; and the capacitors are not storing energy for the machinery to use, because the energy they exchange is returned to the circuit on every half cycle rather than delivered to the shaft. What the reduced line current buys is real: lower losses in the conductors, released capacity in the feeder and transformer, and relief from utility charges based on demand or on power factor. Correction is sized rather than maximized, because overcorrecting drives the power factor leading and brings its own problems.

Q39. Answer: D

Always locate hidden utilities before drilling. Drilling into existing wiring can cause electrocution, fires, or arc flash. Drilling into a gas line creates explosion risk. Use stud finders, electronic pipe/cable locators, or thermal cameras to identify hidden utilities. Never assume a wall is clear.

Q40. Answer: A

Reverse any two phases = reverse rotation. Three-phase motor direction is determined by phase sequence. Swapping any two of the three power conductors at the motor terminals (or disconnect) reverses the rotating magnetic field and thus the motor direction.

Q41. Answer: D

Lone worker rules require a documented check-in or monitoring system. Working alone in an electrical room creates risk - if the worker is injured, there is no immediate help. The duty comes from provincial OH&S regulation, not from a CSA standard. WorkSafeBC's OHS Regulation requires the employer to develop and implement a written procedure for checking the well-being of a worker assigned to work alone or in isolation, stating the time interval between checks and what to do if the worker cannot be contacted, with both the worker and the person doing the checking trained in that procedure. Alberta's OHS Code Part 28 requires an effective communication system - radio, landline or cellular telephone, or other effective electronic means - with contact at intervals appropriate to the hazard, and visits to the worker where electronic contact is not practicable.

Q42. Answer: D

CEC Rule 14-104: the rating of the overcurrent device must not exceed the ampacity of the conductor it protects. On a 20A circuit that means No. 12 AWG copper as a minimum; No. 14 AWG copper is a 15A conductor and must not be run to a 20A breaker, because the wire can overheat and fail before the breaker trips — a fire hazard. The conductor, not the device, is the thing being protected. The separate Rule 14-100 is the one that requires each ungrounded conductor to have an overcurrent device where it receives its supply and at each point its size decreases.

Q43. Answer: B

Table 53 gives 450 mm of cover for a direct-buried armoured cable at 750 V or less outside vehicular areas, and that may be cut by 150 mm where mechanical protection is placed in the trench over the installation. The table has three rows and the row matters as much as the number. Cable with no metal sheath or armour — NMWU and USEI90 are the examples given — takes 600 mm outside vehicular areas and 900 mm under them. Cable with a metal sheath or armour, such as TECK90 or ACWU, takes 450 mm and 600 mm, and a raceway such as rigid PVC or DB2 conduit takes the same 450 mm and 600 mm. So armour buys 150 mm of depth against a bare direct-buried cable, and the protection allowance can buy 150 mm more, putting a protected armoured cable as shallow as 300 mm. Reading the wrong answers: armour answers mechanical damage but does not excuse a cable from a minimum cover, and a run laid just under the sod meets a spade on the first gardening weekend; the 900 mm figure belongs to the non-armoured row under vehicular traffic and is not a universal depth; and 150 mm is below every row of the table. Cover is measured from finished grade to the top surface of the cable or raceway, and every figure rises for installations over 750 V.

Q44. Answer: B

Real power P = V × I × PF = 240 × 20 × 0.75 = 3,600W. Apparent power (VA) = V × I = 240 × 20 = 4,800 VA. Real power (W) = apparent power × power factor. With PF < 1 (inductive or capacitive loads), real power is less than apparent power. The difference is reactive power (VAR).

Q45. Answer: C

Firestop required at all fire-rated penetrations. EMT itself does not restore the fire rating of a wall. CEC Rule 2-128 requires that where a fire separation is pierced by a raceway or cable, any openings around the raceway or cable be properly closed or sealed in compliance with the National Building Code of Canada. An approved firestop system is a tested assembly — sealant, wrap, collar, mortar, putty pad or mineral-wool packing — qualified by the CAN/ULC-S115 fire test to a rating not less than the rating required for that separation. Intumescent sealants and firestop collars are two examples, not the whole family.

Q46. Answer: A

A fault between a consumer's service conductor and the metal raceway around it is on the supply side of the service box, so nothing in the building is upstream of it to clear it. Clearing falls instead to the supply authority's overcurrent device out in its distribution system, and those devices are set so that a great deal of energy is delivered into the fault before they operate. That is why the code asks more of a service raceway than of a feeder raceway: Technical Safety BC's information bulletin on services and service equipment gives exactly this reason for Rule 10-604(2), which requires supplemental bonding connections on cable armour and metal raceway used for a consumer's service that are not required for feeders and branch circuits. In practice that means bonding bushings rather than reliance on a locknut in a knockout - at one end where the raceway contains a bonding conductor, and at both ends where the raceway itself is being used as the bonding conductor. The same worry about conductors that have no effective overcurrent protection runs through the rest of the service rules, which is why service conductors are kept outside the building as far as practicable and the service box is placed close to where they enter it. Reading the wrong answers: the main breaker is not slow here, it is on the load side of the fault and does not see it at all; the raceway is not the grounding electrode, which is the connection to earth and is not the fault-current path; and service conductors sit at the same voltage as the feeders they go on to supply, so voltage is not what makes the difference.

Q47. Answer: B

Live work is permitted only where disconnecting is not reasonably possible, where the equipment is rated 600 volts or less and disconnecting would create a greater hazard than proceeding, or where the work is nothing but diagnostic testing. The regulation sets the default first: for work on or near energized exposed parts, the power supply is to be disconnected, locked out of service and tagged before the work begins and kept that way while it continues, hazardous stored electrical energy is to be discharged or contained, and the worker is to verify both of those before starting. Those three circumstances are the only ones that displace that default. Meeting one of them opens the door but does not finish the job: only a worker holding the electrician certificate of qualification may do the work, written measures and procedures have to be established and given to the worker and explained, and the worker has to use mats, shields or other protective devices, including personal protective equipment, adequate to protect against electrical shock and burn. At a nominal 300 volts or more, a competent worker equipped for rescue including cardiopulmonary resuscitation must be stationed where the live work can be seen, and that duty is lifted only where the work is diagnostic testing alone. Note the trap in that last point: taking readings with a meter is itself work on energized equipment, not a way around the rule. Reading the wrong answers: the certificate and the rubber gloves worn inside leather protectors are both genuine duties, but they are conditions on how live work is carried out, never the reason it may be carried out; the supervisor's authorization governs who may enter a room or enclosure containing exposed energized parts, which is a separate duty again; and an arc flash label with matching arc-rated clothing is sound practice out of the workplace electrical safety standard, but no label licenses live work that could have been avoided by opening a disconnect.

Q48. Answer: B

Ground fault protection guards the equipment, not the person. A line-to-ground fault on a large service can settle into a low-level arc that draws far less current than the main breaker's trip setting, so the breaker sits there while the arc burns through busbar and enclosure. Ground fault protection senses the current leaving on the ground path and opens the main before that happens. It is not a Class A GFCI: a GFCI trips at a few milliamperes to keep a person alive, while ground fault protection is set orders of magnitude higher, to save the switchgear. Whether it is required depends on the ampere rating of the service and on the voltage to ground, and the CEC calls for it in one case the US NEC does not, so read the current thresholds from the code edition in force where you work.

Q49. Answer: B

The replacement has to be protected by a ground fault circuit interrupter of the Class A type, because Rule 26-704 reaches a receptacle of 5-15R or 5-20R configuration installed within 1.5 m of a wash basin, bathtub or shower stall, and a receptacle beside the basin sits well inside that distance whether the outlet is new or long established. The Electrical Safety Authority puts this exact question to itself in its bulletin on replacements and alterations in dwelling units, and answers it yes. The confusion that bulletin exists to settle is the part worth carrying onto a job, because the same document treats arc fault protection the other way round: replace or relocate a panel and leave the existing branch circuit wiring alone, and arc fault protection is not required to be added; swap a receptacle that sits within 1.5 m of the basin, and ground fault protection is. One requirement follows the outlet you are putting your hands on, the other follows branch circuit wiring you are not touching. Reading the wrong answers: a plain replacement leaves the person standing at a wet basin relying on the branch circuit overcurrent device, which will not move for a current that is already lethal; the protection may come from a device at the outlet or from one upstream of it, and a ground fault receptacle at this location is the ordinary way it is done, so nothing pushes the job back to the panel; and arc fault protection watches the waveform for the signature of arcing, which makes it fire protection under a separate rule, and it does not respond to current leaving the circuit through a person. A ground fault device also has to be tested with its own test button on a schedule, because the electronics can fail with the receptacle still delivering power.

Q50. Answer: B

Zero resistance on a coil = shorted windings. A coil is made of many turns of fine wire — it should have measurable resistance (typically 20–500Ω depending on size). Zero ohms means the insulation between turns has broken down (short). The coil will draw excessive current, blow fuses, and produce heat until it burns open.